
LECTURE NOTES

Lecture 3 - Feedforward Networks

AI506 — Advanced Machine Learning

Simon Holm
February, 2026



DEPARTMENT OF MATHEMATICS
AND COMPUTER SCIENCE

Contents

Activation functions	3
Common output units	3

Activation functions

- softmax
- ELU
- SELU
- softplus
- ReLU – $x = \max(0, x_i)$
- sigmoid
- Tanh

Common output units

- **Linear units:** no non-linearity (used for regression)
- **Sigmoid units:** each individual output is between 0 and 1 (used for many-out-of- K classification)
 - can assign multiple classes
- **Softmax units:** each individual output is between 0 and 1 and all outputs together sum to 1 (e.g. used for one-out-of- K classification)
 - can only choose one class